

J Gaurav Varma

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SUMMARY

Software Engineer passionate about designing scalable distributed systems and building production-grade full-stack applications from architecture to deployment. Experienced in developing AI-powered platforms, real-time applications, and high-throughput backend services using React, Next.js, Node.js, FastAPI, PostgreSQL, Redis, Kafka, Docker, and AWS. Strong foundation in system design, performance engineering, asynchronous processing, caching, and cloud-native architectures, with a focus on building resilient, maintainable, and scalable software.

SKILLS

Languages: JavaScript (ES6+), TypeScript, Python, SQL

Frontend: React.js, Next.js, Redux, TanStack Query, Tailwind CSS

Backend: Node.js, Express.js, FastAPI, REST APIs, WebSockets, JWT Authentication

AI Engineering: RAG, OpenAI API, pgvector, Vector Search, Prompt Engineering, Streaming Responses

Databases & Storage: PostgreSQL, MongoDB, Redis, AWS S3

Distributed Systems: Kafka, Event-Driven Architecture, Background Workers, Asynchronous Processing, Distributed Locking

Cloud & DevOps : Docker, AWS (EC2, S3, ECS, Lambda, Cognito, RDS), Nginx, GitHub Actions, CI/CD

Performance Engineering : Caching, Code Splitting, Lazy Loading, Debouncing, Image Optimization, CDN, Lighthouse Optimization

Observability & Monitoring: Grafana, Prometheus, Loki, Logging, Monitoring

System Design: Load Balancing, Rate Limiting, Horizontal Scaling, Vertical Scaling, Database Indexing, Replication, Sharding, API Design

EXPERIENCE

[TRADEBRAINS.IN](#) | Software Engineer

October 2025 - May 2026

- Contributed to a **stock research and analysis platform** where I built an AI chatbot with **context-aware session memory** and **streaming responses**, **reducing** AI reply **latency** from 5-7 seconds to **~1.5 seconds** and capable of analysis of stocks and its past performance. Improved application performance by **25% (Lighthouse score: 62 to 84)** by implementing **route based code splitting**, **optimizing LCP through WebP image compression**, and reducing Total Blocking Time by deferring non critical scripts.
- Contributed to a **Crypto Trading Terminal** by building a **real time market data layer** using **WebSockets** and a **custom API client** to **synchronize live order books**, market feeds, and user balances. Implemented secure session management using **JWT auth**. Improved trader workflow by providing an integrated platform for **strategy creation**, **automated execution**, and **real time PnL monitoring**.

[SIRPI.IO](#)

December 2024 - September 2025

- Architected** and developed a **multi-tenant AI Chatbot as a service** using Next, FastAPI, PostgreSQL (**pgvector**), **Redis**, **Kafka**, **AWS S3** and **EC2**. Designed a scalable **RAG architecture** with **asynchronous crawling**, **document ingestion**, **embedding generation**, and **semantic retrieval pipelines**. Leveraged **Kafka based workers**, **Redis caching**, **distributed locking**, and **stateless services** to **minimize request latency**, **maximize throughput**, and support **horizontal scaling** while integrating **third-party LLM APIs** for context-aware responses.
- Redesigned** and **optimized** a **high traffic visa verification portal** serving **2,500+ international seminar attendees** after identifying **Google Sheets API rate-limiting bottleneck** through **K6 load testing**. **Re architected the lookup workflow** by introducing a **Redis backed caching layer** with **scheduled synchronization**, **eliminating per request external API calls** and **preventing 403 rate-limit failures**. **Collaborated with stakeholders** to define **scalability requirements**, engineering a solution capable of handling **1,000+ concurrent users** while **significantly reducing lookup latency**
- Built a geospatial analytics platform, implementing advanced interactive **dashboards** and **real time data visualization** using **Apache ECharts**, while also **refactoring legacy code** to improve maintainability and performance.
- Co-developed a large scale ERP Platform**. Improved frontend architecture by introducing **TanStack Query**, standardizing API interactions through **centralized data fetching**, **caching**, **automatic retries**, and **background synchronization**, significantly **reducing redundant network requests**, boilerplate code, and improving maintainability. **Optimized invoice generation** by migrating PDF rendering from the browser to the backend, **eliminating UI lag** and delivering a faster, more responsive user experience .

[Goltdev Technologies](#)

Jul 2024 - Aug 2024

- Built a centralized **Admin Dashboard** and **Activity Log Management System** to track and manage application-wide user actions and system events for auditing, debugging, and operational monitoring. Integrated **Grafana**, **Prometheus**, and **Loki** to enable real-time observability, log aggregation, and performance monitoring, improving troubleshooting efficiency and minimizing application downtime.

Personal Projects

Distributed RAG Platform for High-Traffic AI Assistants:

- Architected and developing a distributed **RAG platform** capable of serving high-concurrency AI chat sessions using **FastAPI**, **PostgreSQL (pgvector)**, **Redis**, **Kafka**, **AWS S3**, and **OpenAI APIs**. Designed asynchronous document ingestion and vectorization pipelines using **Kafka-based job queues**, enabling large document uploads, embedding generation, and indexing without blocking user requests. Engineered separate execution paths for low-latency queries and long-running AI tasks

Distributed Real-Time Chat Platform:

- Architected a horizontally scalable real-time chat platform using **FastAPI**, **WebSockets**, **Kafka**, **Redis Pub/Sub**, **MySQL**, and **MinIO** following a microservices architecture. Designed an **event-driven messaging pipeline** where WebSocket servers publish chat events to Kafka for asynchronous persistence while Redis Pub/Sub synchronizes messages across distributed WebSocket instances. Built service discovery, API Gateway, centralized configuration, JWT authentication, OpenTelemetry observability, and containerized deployment using Docker.